QUESTION 1: CALCULATED NUMERIC

5 out of 5 points

Consider a signal x(t) whose Fourier transform is given by $X(f)=\frac{1}{f^2}$. If the Fourier transform of x(-4t) can be written as $k\frac{1}{f^2}$, what is the value of the constant k?

Given Answer: 4

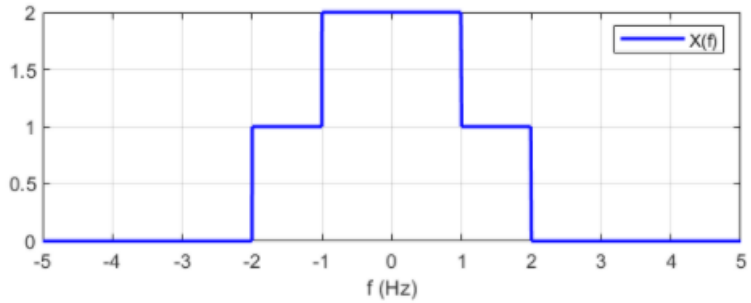
Correct Answer: 4

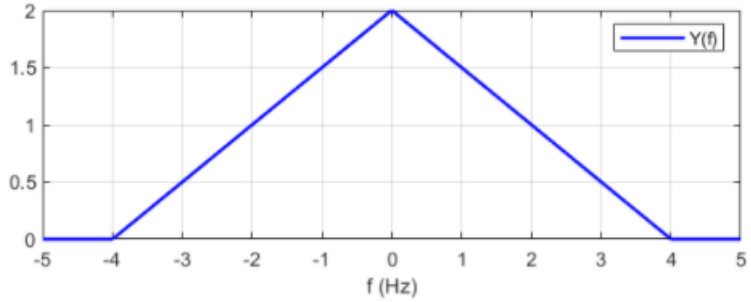
Answer range +/- 0 (4 - 4)

QUESTION 2: CALCULATED NUMERIC

5 out of 5 points

Frequency Division Multiplexing (FDM) is applied on the signals x(t) and y(t) whose spectra are shown below. What must be the minimum separation between the carrier frequencies in order to avoid interference?





Given Answer: 6

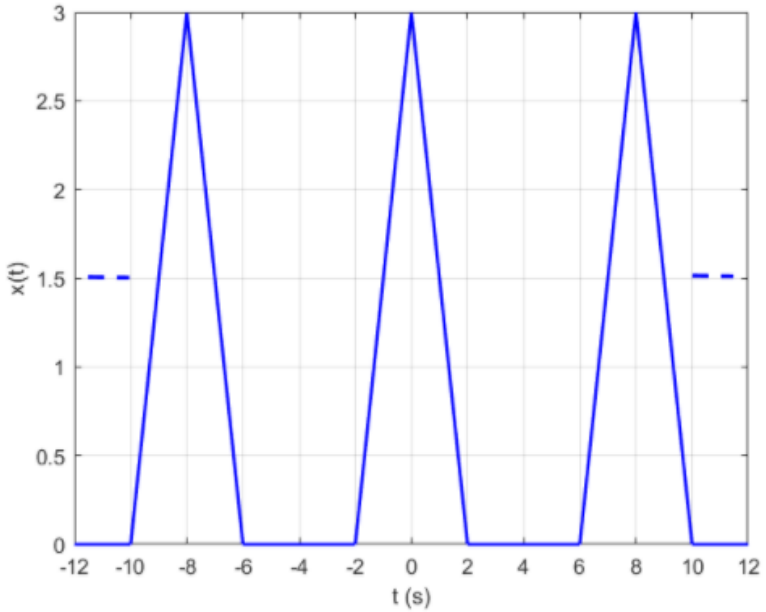
Correct Answer: 6

Answer range +/- 0 (6 - 6)

QUESTION 3: CALCULATED NUMERIC

10 out of 10 points

Consider the periodic signal shown below. If the Fourier series coefficients can be written as $X[k] = a \times \text{sinc}^2(bk)$, what is the value of the constant a?



Given Answer: 0.75

Correct Answer: 0.75

Answer range +/- 0 (0.75 - 0.75)

QUESTION 4: CALCULATED FORMULA

5 out of 5 points

Consider the integral $I = \int_{-\infty}^{+\infty} \text{sinc}\left(\frac{t-a}{b}\right) dt$. What is the value of I if a=12 and b=7?

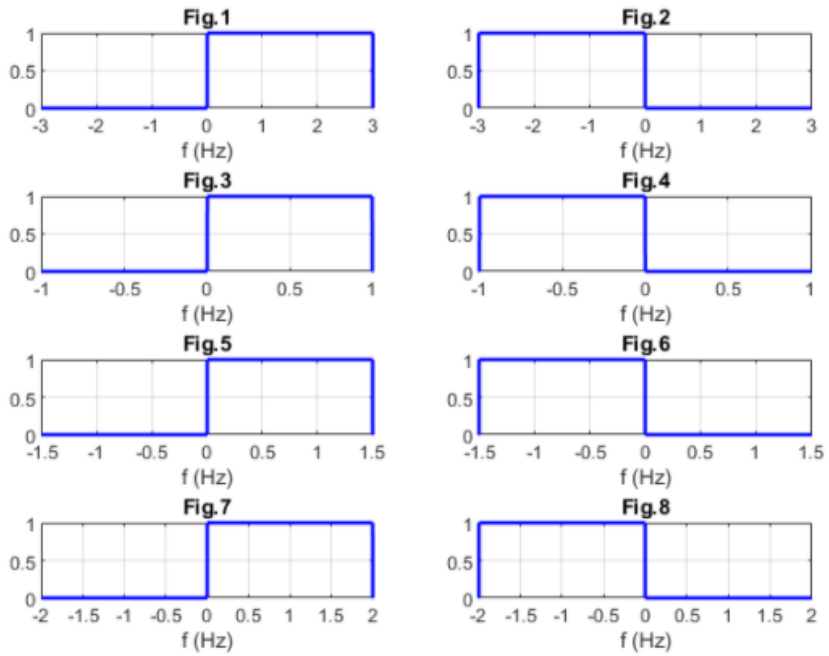
Given Answer: 7

Correct Answer: 7

QUESTION 5: MULTIPLE CHOICE

10 out of 10 points

Consider the signals $x(t) = 2\text{sinc}(2t)$ and $y(t) = 3\text{sinc}(3t)e^{-j\pi 3t}$. Which one of the below figures corresponds to the Fourier transform of the signal x(t)*y(t)? [note that * stands for convolution]



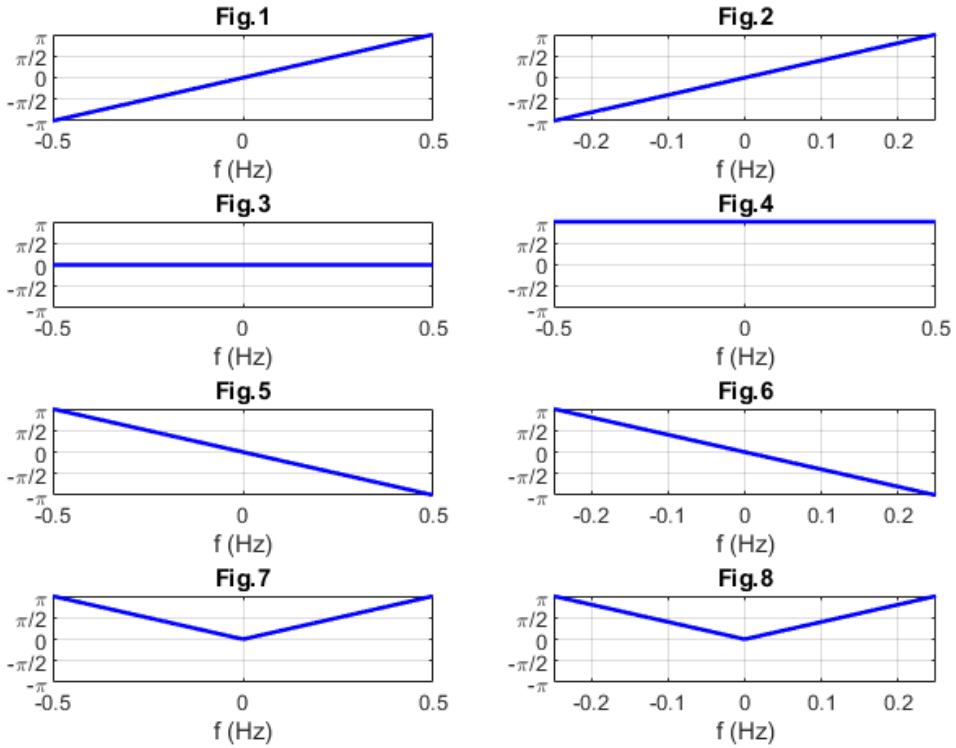
Given Answer: Fig. 4

Correct Answer: Fig. 4

QUESTION 6: MULTIPLE CHOICE

10 out of 10 points

Which one of the below figures corresponds to the phase spectrum of the signal tri(t-1)?



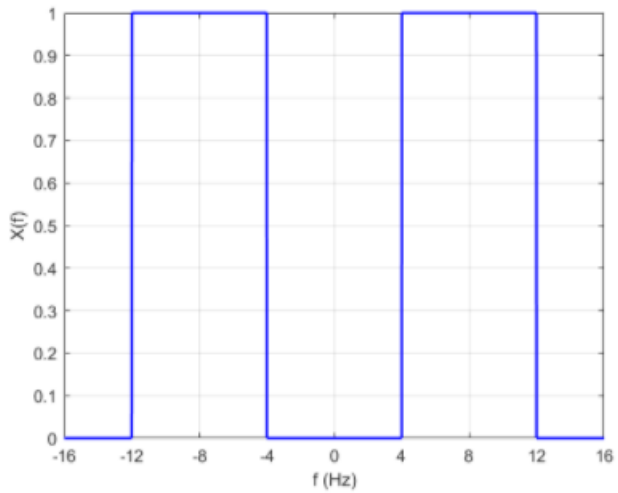
Given Answer: Fig. 5

Correct Answer: Fig. 5

QUESTION 7: MULTIPLE CHOICE

10 out of 10 points

Consider the signal whose spectrum is shown below:



If the signal can be written as $x(t) = k \times \text{sinc}(at)\cos(2\pi bt)$, which row of the below table contains the correct values of the constants k, a and b?

Row #	k	a	b
Row 1	2	2	8
Row 2	4	2	8
Row 3	2	2	10
Row 4	4	2	10
Row 5	4	4	8
Row 6	8	4	8
Row 7	4	4	10
Row 8	8	4	10
Row 9	8	8	8
Row 10	16	8	8
Row 11	8	8	10
Row 12	16	8	10

Given Answer: Row 10

Correct Answer: Row 10

QUESTION 8: MULTIPLE CHOICE

5 out of 5 points

Consider the following signals:

$x(t) = e^{-2t}u(t)$

$y(t) = e^{-3t}u(t)$

$z(t) = e^{-4t}u(t)$

$w(t) = e^{-5t}u(t)$

Which one of the above signals has the broadest spectrum?

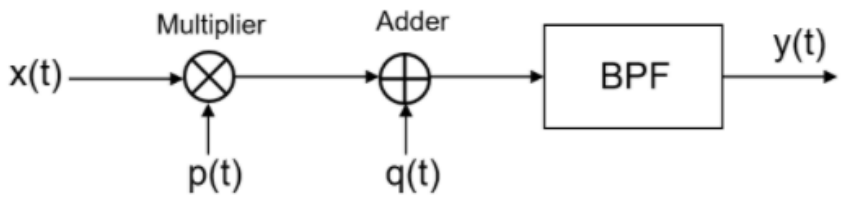
Given Answer: w(t)

Correct Answer: w(t)

QUESTION 9: MULTIPLE CHOICE

15 out of 15 points

Consider the following system:

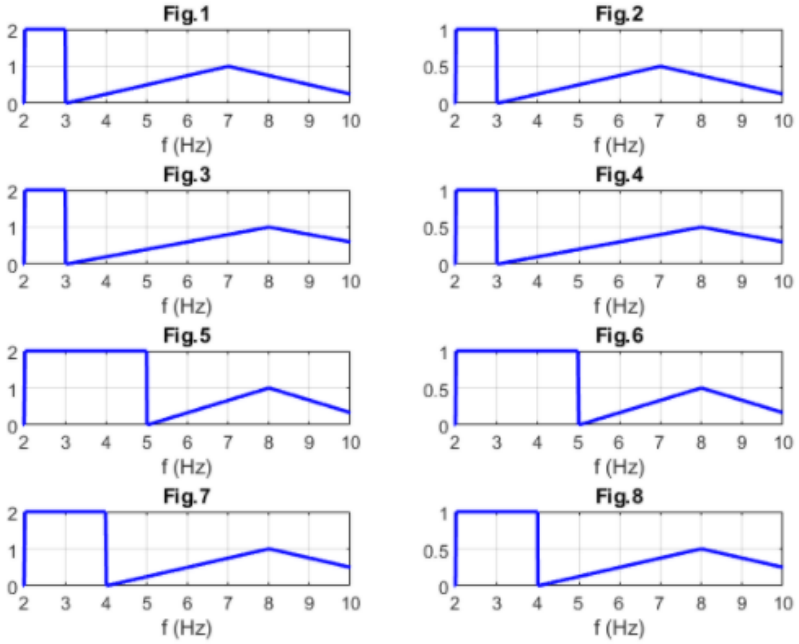


where:

$x(t) = 4\text{sinc}^2(4t)$; $p(t) = \cos(16\pi t)$; $q(t) = 8\text{sinc}(8t)$

and the band-pass-filter (BPF) is ideal with gain 1, lower cutoff frequency 2 Hz and upper cutoff frequency 10 Hz.

Which one of the below figures corresponds to Y(f) in the frequency band [2 10] Hz?



Given Answer: Fig. 8

Correct Answer: Fig. 8

QUESTION 10: CALCULATED NUMERIC

5 out of 5 points

Consider the signal x(t)=cos(200πt) and denote its Fourier transform by X(f). What is the value of the following integral?

$$I = \int_{-\infty}^{+\infty} X(f)\,df$$

Given Answer: 1

Correct Answer: 1

Answer range +/- 0 (1 - 1)

QUESTION 11: MULTIPLE CHOICE

10 out of 10 points

The signal x(t)=4comb(4t) is applied to the input of an ideal low-pass filter with gain 3 and cutoff frequency 6 Hz. If the output signal can be written as y(t)=a+b×cos(2πct), which row of the below table contains the correct values of the constants a, b and c?

Row #	a	b	c
Row 1	3	3	2
Row 2	3	6	2
Row 3	6	6	2
Row 4	6	12	2
Row 5	3	3	4
Row 6	3	6	4
Row 7	12	12	4
Row 8	12	24	4
Row 9	3	3	8
Row 10	3	6	8
Row 11	24	24	8
Row 12	24	48	8

Given Answer: Row 8

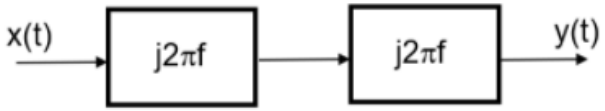
Correct Answer: Row 8

QUESTION 12: MULTIPLE CHOICE

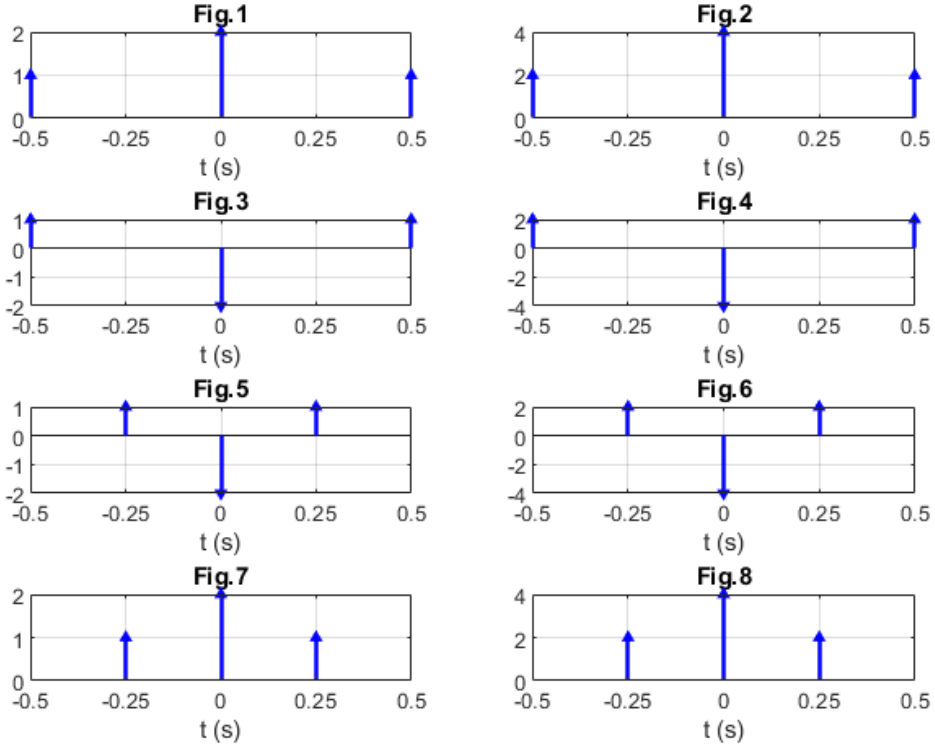
10 out of 10 points

Copy of

Consider the system shown below:



If the input signal is given by x(t)=tri(2t), which one of the below figures corresponds to the output signal y(t)?



Given Answer: Fig. 4

Correct Answer: Fig. 4